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ORIGINAL ARTICLE

Evaluation of a fall-prevention program in older people after femoral neck fracture: a one-year follow-up

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Abstract
Summary A randomized, controlled fall-prevention study including 199 patients operated on for femoral neck fracture reduced inpatient falls and injuries. No statistically significant effects of the intervention program could be detected after discharge. It seems that fall-prevention must be part of everyday life in fall-prone old people.
Introduction This study evaluates whether a postoperative multidisciplinary, multifactorial fall-prevention program performed by a geriatric team that reduced inpatient falls and injuries had any continuing effect after discharge. The intervention consisted of staff education, systematic assessment and treatment of fall risk factors and vitamin D and calcium supplementation.
Methods The randomized, controlled trial with a one-year follow-up at Umeå University Hospital, Sweden, included 199 patients operated on for femoral neck fracture, aged ≥70 years.
Results After one year 44 participants had fallen 138 times in the intervention group compared with 55 participants and 191 falls in the control group. The crude postoperative fall incidence was 4.16/1,000 days in the intervention group vs. 6.43/1,000 days in the control group. The incidence rate ratio was 0.64 (95% CI: 0.40–1.02, p=0.063). Seven new fractures occurred in the intervention group and 11 in the control group.
Conclusion A team applying comprehensive geriatric assessment and rehabilitation, including prevention and treatment of fall-risk factors, reduced inpatient falls and injuries, but no statistically significant effects of the program could be detected after discharge. It seems that fall-prevention must be part of everyday life in fall-prone elderly.

Keywords Accidental falls · Geriatric care · Hip fracture · Intervention · Randomized control trial

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Introduction

Hip fracture is a common and serious incident that occurs among older people. The growing incidence is and will continue to be a major health problem for society [1]. The annual hip fracture incidence in Sweden is 18,000 [2]. Only 50% of those suffering hip fracture regain their previous ability regarding walking and activities of daily living (ADL), one in four have to move to an institution. The mortality 5 years after fracture is 20% above the expected level, and most deaths occur within the first 6 months [3]. Almost all hip fractures occur as a result of a fall [4, 5], and many will fall again soon after sustaining the fracture [6, 7]. There is a 2.5-fold increased risk of sustaining a subsequent fracture after a first hip fracture [8]. A population-based

study among people aged 85 years or older showed that 21% of those with a hip fracture had suffered at least two such fractures [9].

Previous research has identified several fall-risk factors, such as co-morbidity, functional disability, previous falls, use of drugs, aging and among the oldest old, male sex [10–12]. Fall prevention measures, both single and multifactorial interventions, have been successful among older people living in the community while only multifactorial interventions have reduced falls and injuries in older people in institutional care [11, 13, 14].

We have recently shown that the number of falls, fallers and fall-related injuries can be prevented during hospitalization after a femoral neck fracture by implementing a multidisciplinary, multifactorial rehabilitation program, which includes prevention and treatment of fall-risk factors [15]. The participants were also treated with calcium and vitamin D, which might prevent both falls and fractures [16, 17]. This study showed a significant reduction in falls in the intervention group, especially among the participants suffering from dementia, who benefited most regarding prevention of postoperative falls.

The aim of the present study was to evaluate whether a postoperative multidisciplinary, multifactorial fall-prevention program performed by a geriatric team that reduced inpatient falls and injuries had any continuing effect after discharge.

Methods

Sample

This study included participants with femoral neck fractures aged 70 years or older admitted consecutively to the orthopedic department at the Umeå University Hospital in Umeå, Sweden, between May 2000 and December 2002. The one-year follow-up ended in December 2003. The following exclusion criteria were used: rheumatoid arthritis, severe hip osteoarthritis and pathological fracture, because of the operation methods planned to be use in the study. The anesthesiologist decided whether the planned operation according to the study protocol was appropriate for the patient. Patients with severe renal failure were excluded by the anesthesiologist because of their morbidity. Those bedridden before the fracture were also excluded.

Procedure

In the emergency room the patients were asked both in writing and orally if they were willing to participate in the study. The next of kin were also asked in the case of patients with cognitive impairment. Two hundred and fifty-eight patients met the inclusion criteria, 11 patients declined to participate and 48 patients were excluded because they had sustained the fracture in the hospital (21/258, 8%) or the inclusion routines had failed (Fig. 1). These 59 patients were more likely to be men ($p=0.033$), and living in their own house/apartment ($p=0.009$) but there was no difference in age ($p=0.354$) compared to those who participated. The remaining 199 patients consented to be part of the study and they all received the same preoperative treatment.

Randomization

Participants were randomly assigned by someone who was not involved in the study, to postoperative care on a ward in the geriatric rehabilitation department with a special intervention program (intervention group) or to the orthopedic department with conventional care (control group). All participants received lots sequentially numbered and stratified according to operation method in an opaque, sealed envelope at the emergency room. The envelope was not opened until immediately before surgery to ensure similar preoperative treatment. Those randomized to the control group who needed a longer rehabilitation period were transferred to a geriatric rehabilitation ward ($n=40$), but not to the same ward where the intervention program was implemented.

The staff on both the control and the intervention ward was not aware of the follow-up for falls in the present study.

The intervention

The intervention ward was in a geriatric department specializing in geriatric orthopedic patients, the staff worked in teams to apply comprehensive geriatric assessments and rehabilitation [18, 19]. Active prevention, detection and treatment of risk factors for falls such as delirium, pain and infections were systematically implemented. Those with postoperative delirium were carefully assessed for potential precipitating factors, since delirium by definition always has an underlying cause [20]. All patients were prescribed 1000 mg calcium and 800 IU/d of vitamin D. The staffing on the intervention ward was 1.07 nurses/aids per bed.

About two weeks after discharge the participants in the intervention group received a telephone call from a physiotherapist or an occupational therapist to ask if there was anything they needed attendance with.

The participants living in the community in need of further rehabilitation were offered day-care rehabilitation at the geriatric department or contact with the physiotherapist and occupational therapist working with the family practitioner. Participants in institutional care who needed rehabilitation after discharge were assessed by therapists

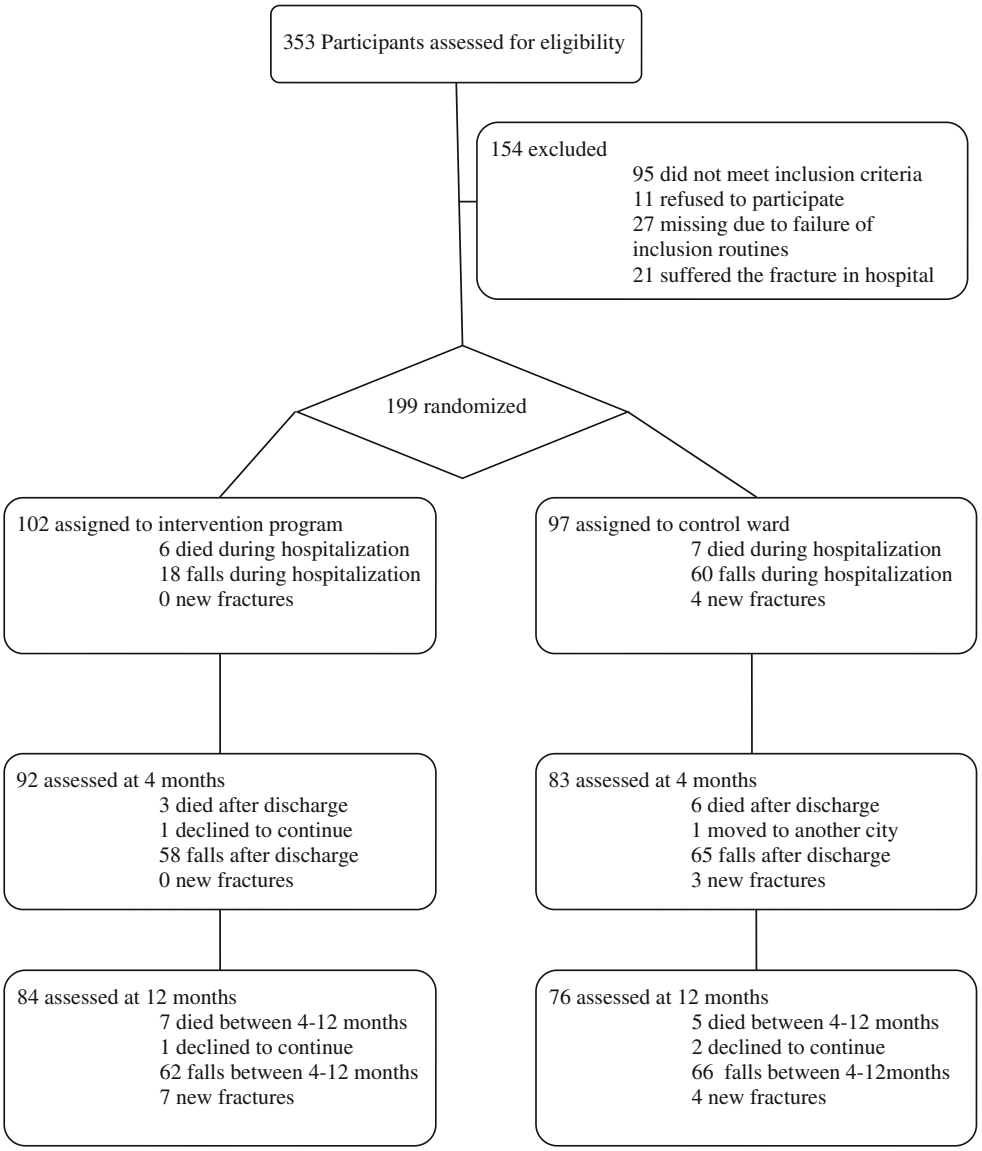


Fig. 1 Diagram shows the results of all 353 patients in the study

employed by the community. Home training programs were offered to those who were not assigned to any other kind of rehabilitation.

The participants in the intervention group met a geriatric specialist at a 4-month visit, either at the geriatric outpatient clinic or if the participant's condition was poor the doctor made a home visit. The doctor checked for fall-risk factors, such as low blood pressure, inappropriate medication and other complications. The doctor also gave a prescription of calcium and vitamin D if it was not prescribed at discharge. A nurse and a physiotherapist or an occupational therapist assessed the need for rehabilitation, need for assistive devices, modification of the environment and nutritional problems in the intervention group at the 4-month follow-up.

The control ward was in a specialist orthopedic department following the normal postoperative routines used at the orthopedic department. A geriatric ward, specializing in general geriatric patients was used for those who needed longer in-hospital rehabilitation. The staffing on the orthopedic ward was 1.01 nurses/aids per bed and 1.07 on the geriatric control ward. The content of both the intervention program and the conventional care has been presented in detail in a recently published article [15].

Data collection

Two registered nurses were each employed half-time in the project. The one from the orthopedic department performed the assessments in the intervention group and the other from the geriatric rehabilitation department carried out the same assessments in the control group. The two nurses, accompanied by either a physiotherapist or an occupational therapist, collected data during home visits at 4 and 12 months.

Medical and social data, including morbidity and mortality and the occurrence of falls, were collected and registered from the medical and nursing records and from interviews with participants, relatives and staff, on admission and during follow-ups. A fall was defined as an incident when the participant unintentionally came to rest on the floor or ground and included for example syncopal falls. All nurses in hospitals and institutional care are obliged to document falls in the records. Observation time was calculated from the day of admission until the day of final follow-up. To give an example; if the participant had died or declined between 4 and 12-month follow-up, only observation days up to 4 months were calculated.

Three to five days after surgery, participants were assessed and interviewed regarding their cognitive status using the Mini Mental State Examination (MMSE) [21]. The modified Organic Brain Syndrome Scale (OBS) [22] was used to assess cognitive, perceptual, emotional and personality characteristics, as well as fluctuations in clinical states. The patients were also screened for depression using the Geriatric Depression Scale (GDS-15) [23]. Depression during hospitalization was diagnosed on the basis of previously diagnosed depression with ongoing treatment with antidepressants and assessments with the GDS and OBS scales.

The participants' vision and hearing were assessed by their ability to read three-millimeter block letters with or without glasses, and their ability to hear a normal speaking voice from a distance of one meter. The functional status of ADL performance prior to the fracture was measured using the staircase of ADL [24]. A high score indicates greater ADL dependence. Walking ability prior to the fracture was registered according to one item in the Swedish version of the clinical outcome variables (S-COVS) [25, 26]. With normal function the person has a high score.

The same assessments were performed at the 4- and 12-month follow-ups and in addition the participants were assessed using a modified version of the chair stand test [27] and Berg´s balance scale [28]. The chair stand test was performed by testing the participant's ability to rise once from a straight-backed chair with the help of the arms.

A geriatrician, unaware of study-group allocation, analyzed all assessments and documentation after the study was finished, to decide whether the participants fulfilled the DSM-IV criteria [20] for dementia, delirium and depression.

The ethics committee of the faculty of medicine at Umeå University approved the study (§ 00–137).

Statistical analysis

Student's t-test, Pearson's chi-square test, Fisher's exact test and the Mann–Whitney U test were performed to analyze group differences regarding basic characteristics and mortality. Outcomes were analyzed on an intention-to-treat basis.

The risk of falls between intervention and control groups was compared in various ways. First an unadjusted comparison was made using Pearson's chi-square test and Fisher's exact test regarding the number of participants who fell and fractures. Second, the fall incidence rate was compared between intervention and control groups by calculating the fall incidence rate ratio (IRR) using negative binomial regression (Nbreg), which is a generalization of the Poisson regression model, with adjustment for over-dispersion and observation time.

Baseline characteristics that differed between the intervention and the control groups, corresponding to a p-value <0.15 (depression and dementia, see Table 1) were considered as covariates in the Poisson (Nbreg) regression model. These variables had only marginal effect on the IRR values and are therefore not included in the presentation of the data.

All calculations were carried out using SPSS v 12.01 and STATA 9 statistical software. A p-value <0.05 was considered statistically significant.

Sample size

From earlier studies we projected that 50% of the participants in the control group would fall. With this sample size we calculated that a 20% reduction in falls in the intervention group could be detected with a power of 80%, two tailed α=005.

Results

No significant differences between intervention and control groups could be seen when comparing falls and new fractures between discharge and the 4-month follow-up, nor between the 4− and 12-month follow-up, nor cumulatively at the 12-month follow-up.

Analysis of the data from the 4-month follow-up showed that a total of 26/92 participants in the intervention group had sustained 58 falls and 26/82 participants in the control group had sustained 65 falls. The crude fall-incidence rate was 6.30/1 000 days in the intervention group vs. 9.07/1 000 days in the control group. Using a Poisson regression (Nbreg), adjusting for over-dispersion, the IRR was 0.55 (95% CI: 0.27–1.12, p=0.100) (Table 2). Among participants with dementia, 8 in the intervention group sustained 22 falls and 11 in the control group sustained 38 falls.

Between four and twelve months no further significant differences were detected. IRR was 0.85 (95% CI: 0.48–1.50, p=0.577) (Table 2).

Table 1 Basic characteristics and assessments among participants, intervention- and control group

	Basic characteristics during hospitalization		Basic characteristics at 4 months		Basic characteristics at 12 months	
	Control (n=97)	Intervention (n=102)	Control (n=83)***	Intervention (n=92)	Control (n=76)	Intervention (n=84)
Sociodemographic						
Mean age, mean ± SD	82.0±5.9	82.3±6.6	81.9±5.6	82.3±6.8	81.8±5.5	82.0±6.6
Females	74	74	64	68	60	62
Independent living before /after the fracture	60	66	46	54	36	47
Health and medical problems						
Cancer	14 (92)	15	13	12	12	10
Cardiovascular disease	53 (93)	57 (101)	44	51	40 (75)	47
Dementia	36	28	32	31 (91)	40	32 (83)
Depression	45 (95)*	33	41	49 (91)	35 (73)	43 (82)
Diabetes	17 (95)	23	11 (81)	23	11 (74)	18
Previous hip fracture	14 (96)	16				
Previous wrist fracture	23 (95)	16 (101)				
Stroke	20 (93)	29	13 (82)*	29	15	27
Medications at admission/at follow-up						
Number of drugs, mean ± SD	5.9±3.6	5.8±3.8	7.3±4.3 (82)	7.5±4.2 (90)	7.8±4.8 (68)	8.3±4.2 (83)
Antidepressants	45*	29	34*	25	36*	25
Analgesics	32	30	45	46	47*	38
Benzodiazepines	25	21	27	20	20	16
Calcium/Vitamin-D****	16	12	17**	51	19**	60
Neuroleptics	8	13	7	13	8	11
Sensory impairments						
Impaired hearing	34 (82)	42 (94)	33 (76)	37 (86)	29 (69)	34 (80)
Impaired vision	27 (74)	37 (91)	18 (52)	24 (61)	20 (62)	31 (80)
Functional performance /assessments before and after fracture						
Berg's balance scale, mean+SD			31.1±16 (72)	34.7±15.4 (73)	29.6±17.6 (67)	34.5±16.1 (68)
Geriatric depression scale, mean ± SD	4.5±3.5 (68)	5.2±3.6 (81)	3.4±2.8* (72)	4.4±3.1 (78)	3.6±3.2** (64)	5.2±3.1 (74)
Manage chair stand test with arms			53 (71)	64 (77)	33 (68)	40 (75)
Mini mental state examination, mean±SD	15.7±9.1 (90)	17.4±8.2 (93)	19.1±8.6 (82)	19.1±8.7 (89)	17.9±9.3	19.5±8.6 (83)
SCOVS, need of assistance median (Q1,Q3)	5.5 (5–7) (94)	6 (5–7) (101)	5 (3–6) (82)	5 (3–6)	5 (2–6)	5 (3–6)
Staircase of ADL, median (Q1,Q3)	5 (0.25–7) (88)	5 (1–7.75) (92)	7 (4.75–9)	6 (3–9)	6 (4.25–9)	5 (3–9)

*p<0,05 **p<0,005 ***one person missing at four months ****At 4 months 9/17 and 7/51 had 400 IU/d of vitamin D and 500 mg calcium. At 12 months 8/19 and 8/60 had 400 IU/d of vitamin D and 500 mg calcium
If details are not given for the complete group, the number of subjects is given in parentheses

Between admission and the 12-month follow-up 44 participants in the intervention group sustained 138 falls (range 1–11) and 55 participants in the control group sustained 191 falls (range 1–31). The crude fall-incidence rate was 4.16/1 000 days in the intervention group vs. 6.43/1,000 days in the control group. IRR was 0.64 (95% CI: 0.40–1.02, p=0.063) (Table 2). Among participants with dementia (diagnosed at baseline) 12 sustained 41 falls in the intervention group and 22 sustained 104 falls in the control group during one year (Table 2). The crude fall-incidence rate was 5.16/1 000 days in the intervention group vs. 9.52/1 000 days in the control group. IRR was 0.48 (95% CI: 0.21–1.09, p=0.079).

The new fractures at 4 months were one hip fracture, one nose fracture and one fracture of the scapula. All these fractures occurred in the control group. Between 4 and 12 months four participants in the control group sustained new fractures - one hip fracture, one pelvic fracture, one proximal humerus fracture and one vertebral fracture. In the intervention group, seven participants suffered two pelvic fractures, one proximal humerus fracture, three wrist fractures, one rib fracture, one proximal tibia fracture, one dens fracture and one face fracture. Three out of seven participants in the intervention group sustained two fractures each. The differences in fracture rate between control and intervention groups were not significant at the 4− and 12-month follow-ups.

Table 2 Falls at 4- and 12-month follow-up and cumulatively at 12 months

	Between discharge and 4 months		Between 4 and 12 months		Between admission and 12 months	
	Control n=83**	Intervention n=92	Control n=76	Intervention n=84	Control n=97	Intervention n=102
Number of fallers	26 (82)	26	30	31	55	44
Recurrent fallers, ≥2 falls	12	13	14	14	35	29
Total number of falls	65	58	66	62	191	138
Number of fallers among people with dementia	11 (30)	8 (24)	11 (28)	8 (19)	22 (36)	12 (28)
Recurrent fallers among those with dementia	6	5	6	5	14	7
Total number of falls among people with dementia	38	22	32	18	104	41
Number of fallers with fractures due to falls	3	0	4	7	11	7
Number of demented fallers with fractures due to falls	1	0	1	3	5	3
Observation time(days)	7169	9206	18653	20932	29707	33202
Crude fall incidence rate	9.07/1000	6.30/1000	3.54/1000	2.96/1000	6.43/1000	4.16/1000
Incidence rate ratio with 95% CI		IRR 0.55* (0.27–1.12)		IRR 0.85* (0.48–1.50)	IRR 0.64* (0.40–1.02)	

* Negative binomial regression analyses adjusted for over-dispersion and controlled for dementia and depression. **One person missing, declined at 4-month follow-up but was followed up at 12 months
If details are not given for the complete group, the number of subjects is given in parentheses

After one year, 16 out of 102 (16%) participants in the intervention group and 18 out of 97 (19%) participants in the control group had died (p=0.591).

Discussion

This study shows that falls and fractures after a previous hip fracture constitute a major health problem. Half of the participants fell and sustained in total 329 falls; almost one in five of the fallers suffered a new fracture and more than one in six of the participants died during the first year after fracture. We have previously shown that the present multidisciplinary, multi-factorial geriatric rehabilitation program with systematic assessment and treatment of fall-risk factors, active detection and treatment of postoperative complications resulted in fewer participants with falls, a lower total number of falls and fever fall-related injuries during hospitalization [15]. No statistically significant effect on the proportion of fallers, number of falls, fractures or mortality could be detected after discharge from hospital.

Hip fracture patients have a high risk of falling and this is the first fall-prevention study including only such a group, as far as we know. Earlier studies of the effectiveness of coordinated, multidisciplinary, inpatient, rehabilitation in older patients following proximal femoral fracture have been inconclusive [29], although there is a trend towards effectiveness. The outcome has been mortality and living after discharge but none has included falls as an outcome measure. Falls can be prevented in old people living at home and in institutional care [13]. Combining the coordinated, multidisciplinary approach with fall prevention in old hip fracture patients has not previously been done.

In hospital fall-prevention studies with single interventions among older patients in rehabilitation units have not produced significant results [30–32]. Three fall-prevention studies in hospital settings [33–35] with positive outcomes in other participant groups and on acute and sub-acute wards have recently been published. These studies used fall-risk assessment tools to recognize those with a high fall risk. In the present study we used a rehabilitation and care program, including assessment of risk factors for falls, active detection and treatment of complications for everyone, to reduce the falls. The participants included in our study had all suffered hip fractures, compared to a mix of different diagnoses in the earlier studies. Furthermore we studied the participants both acutely and sub-acutely which the other studies did not. None of the other studies followed the participants after discharge, as has been done in the present study.

During the last years the role of vitamin D and calcium has been debated. Today a recommended dose of 700–800 IU/d of vitamin D is needed to prevent falls and fractures [16, 36–38]. A recent comparative metaanalysis suggests that hip fracture risk is reduced only when vitamin D is added oral calcium, and recommend the dose of 1000–1200

mg/d [17]. There is evidence that calcium and vitamin D significantly improve body sway and lower extremity strength, reducing the risk of falling [39]. The intervention group had a high prescription of calcium and vitamin D compared to the control group (66% vs. 22%) at the 4-month follow-up. At one year the prescription rate still differed (71% vs. 25%). The prescribed doses of vitamin D (800 IU/d) and calcium (1000 mg/d) were in some cases lower, the reason for this was not documented. In clinical practice a common reason for reducing the dose is constipation. In this study vitamin D and calcium seemed to have had no effect on reducing falls. No vitamin D concentration was measured in the present study.

A study of vitamin D and risk reduction of falls, in elderly people living in care home accommodation produced a contradictory outcome [40]; the authors speculate whether the case mix in their study was different compared to the previous studies, which might explain why no reduction of fall risk was seen. Our population included one third admitted from institution, and a large proportion with dementia, stroke and depression. In this sample of frail old people with many fall risk factors, as among people in care home, a single intervention (calcium and vitamin D) is probably not enough to reach significant effect on falls.

Antidepressants are also associated with falls [41]. Since there is no significant difference in the number of falls between intervention and control group, even though there is a significant difference when it comes to treatment, this association can not be supported by this study.

Another approach in preventing fractures is to wear hip protectors. There is some doubt about the effectiveness of hip protectors in preventing hip fractures in older people and compliance is poor due to discomfort [42]. There is no data on hip protectors in this study, but we believe there are no significant differences between the groups.

More participants in the intervention group were assigned to geriatric daytime rehabilitation care after discharge but this seems not to have influenced the fall risk during follow-ups.

So why is there no difference between the groups? Are there ***more*** factors to consider? A person able to rise from a chair but who needs help when walking is strongly associated with falling. The intermediate groups when it comes to cognitive function and ADL functions have a high risk of becoming fallers [43]. More participants in the intervention group walked without walking aids indoors and significantly more participants had regained independence in personal ADL [44] at the 12-month follow-up. When comparing the ability to manage the chair stand test with arms and the mean in Berg´s Balance scale a difference can be seen between the groups after a year, although it is not significant. Perhaps a larger proportion of patients in the intervention group remain “at risk of falling”.

Fall prevention was not systematically maintained after discharge in the intervention group. The specially trained personnel, the environment and the degree of attention that the patient received in hospital are important parts of fall prevention; none of these were maintained after discharge.

Finally there are some limitations to this study; the recording of falls among participants living at home could have been done by keeping fall calendars and with a more frequent follow-up. Some falls may have been missed, not many on the ward, but most likely some during follow-up. The fall record after discharge was retrospective; the medical and nursing documentation were assessed in residential care facilities at 4 and 12 months. However, if any falls were missing there would probably be no difference between the groups.

The reduction in falls between discharge and 4 months (IRR 0.55) and between 4 and 12 months (IRR 0.85) did not reach statistical significance. This could be due partly to a too low power of the study.

The circumstances surrounding the falls are not known in the present study. Acute disease or symptoms of disease have been reported to be a contributory factor in approximately 38% of falls in residential care facilities [41, 43]. Among inpatients 45% of falls are associated with delirium [7]. To succeed in preventing falls we believe that predisposing factors, such as gait, balance and chronic diseases must be carefully optimized and followed over time. The precipitating factors, such as acute illness, delirium and acute drug side effects, play an important part and their prevention, detection and treatment should be given high priority; this was not done systematically in the first year of follow-up either in the intervention group or in the control group.

According to this and previous studies, we believe that fall prevention can be successful if maintained continuously and if both predisposing and precipitating factors are included. The patients with dementia have a high risk of falling and they benefited the most when including hospitalization in the present study. Further research as to which interventions are most effective at reducing falls in demented people living at home and in residential care and how to accomplish fall prevention that is successful over time is required.

Conclusion

A team applying comprehensive, geriatric assessment and rehabilitation, including prevention and treatment of fall-risk factors, reduced inpatient falls, but it was not possible to detect any significant effects of the intervention program after discharge from hospital. It seems that fall prevention has to be part of everyday life in fall-prone old people.

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